

# Classifying Danish agricultural (autumn) vegetation types using Sentinel 1- and 2- data



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## Abstract

A deep learning model was developed using satellite data from Sentinel-1 (radar) and Sentinel-2 (optical) to classify Danish autumn vegetation and main crops at field(-tile) level, taking advantage of the unique crop and management records of the Danish Landovervågningsarealer (LOOP-areas). Satellite observations were processed into half-monthly summaries, with both absolute values and temporal differences matrices as input. The model combined a Convolutional Neural Network on the difference matrix and a Guarded Recursive Unit- model on the absolute values. Testing on unseen areas showed reasonable spatial transferability, achieving weighted F1-scores of 0.86 for main crops (10 categories) and 0.69 (7 categories) for autumn vegetation.

## 1 Introduction

Excessive nitrogen loading is causing ecosystem malfunction worldwide (**Akinawo 2023**, **Kaae, Bak, and Damgaard 2025**). Agricultural fertilization is a major source, with 34-78 kg N/ha per year leaching from the root zone to streams and groundwater in five of the six Danish Land Surveillance Catchments (Landovervågningsområder) (**Blicher-Mathiesen et al. 2024**). These catchments have detailed records of crop composition, management, stream chemistry, and, for five areas, soil pore water and groundwater chemistry dating back to the 1990s. These datasets provide valuable long-term information on Danish agricultural practices and their environmental impacts and are used for various environmental modeling tasks (**Blicher-Mathiesen et al. 2024**, **Børgesen et al. 2022**).

Nitrate leaching depends heavily on agricultural practices (**Kumar et al. 2025**,

Li et al. [2023], Zhao et al. [2022]. Winter catch crops (non-harvested crops grown between main crops) can reduce leaching by up to 68% compared to bare fields (Kumar et al. [2025], Vanpoucke et al. [2024]). In Denmark, catch crops must cover 10.7-14.7% of each farm’s fields, with additional “targeted” catch crops mandated (up to 44.2%) in vulnerable areas (SGAV [2025], Landbrugsstyrelsen [2024]). Their impact depends on extent, type, growth, and displaced vegetation. Record detail varies: the LOOP-areas have field-level autumn vegetation data over many years, while most other areas and pre-2019 data are at the farm level, highlighting the need to quantify catch crops independently of registers..

Satellite data provides an independent source of information on field-level vegetation. Sentinel-1 and Sentinel-2 offer widely accessible radar and optical data with 10–60 m resolution and are used for diverse vegetation-modelling tasks, often for main crops (Tufail, Tassinari, and Torreggiani [2025], Campos-Taberner et al. [2019]) or even trees (Immitzer, Vuolo, and Atzberger [2016]), rather than catch crops or broader autumn vegetation. Interest in modelling catch-crop growth has increased, both for biomass estimation (Goffart et al. [2021], Nascimento Bendini et al. [2024]) and type classification (Vanpoucke et al. [2024], Selvaraj et al. [2025], Jensch et al. [2025]).

Vanpoucke et al. used Sentinel-2 optical data in Belgium to classify autumn vegetation into seven groups using a two-step hierarchical 1D-CNN: first detecting presence/absence of winter catch crops, then classifying detected catch crops into three groups and “others” into four groups (including “no crop”) at a thrice-monthly temporal resolution. Using only satellite data, they achieved an F1-score of 0.87–0.89, but did not test spatial transferability (Vanpoucke et al. [2024]).

Selvaraj et al. (Germany) used Sentinel-1 radar data to classify catch crops into six groups using a Random Forest on manually extracted temporal variables. Like Vanpoucke, they applied a two-level hierarchy: first a binary “catch crop” vs. “others” classification, then subgroup classification. Subgroup F1-scores varied widely-from 0.5 for mustard to 0.95 for mixtures in one test year. Spatial transfer testing of the binary step gave F1-scores of 0.67-0.835, while spatial transferability of the subgroup classification was not tested (Selvaraj et al. [2025]).

Jensch et al. found that integrating Sentinel-1 radar and Sentinel-2 optical data for catch crop prescience detection was better than using either alone.(Jensch et al. [2025]).

The unique LOOP-area crop and management records have not yet been used to train a Danish autumn-vegetation classification model based on satellite data. This project aims to develop a model using both Sentinel-1 radar and Sentinel-2 optical time series to classify autumn vegetation independently of

register data. Prior work shows GRUs (Guarded Recursive Units, a type of recurrent neural network) outperform LSTMs (Long-short-term-memory, another type of recurrent network) for 1D crop-time modelling (Tufail, Tassinari, and Torreggiani 2025), and Vanpoucke successfully applied a CNN (Convolutional Neural Network) to optical time series for catch-crop identification (Vanpoucke et al. 2024).

This model will combines both approaches: GRUs and a CNN applied to two distinct representations of the field-level satellite data. The goal is to establish a foundational model for automatic catchment-scale decomposition of autumn vegetation types, supporting analysis of catch-crop effects on nitrogen dynamics. Spatial transferability is essential, so the model will be tested on geographical areas not used in training.

## 2 Methods

### 2.1 Data-acquisition and pre-processing

The satellite data was obtained by requesting data from the Sentinel Hub API (Planet Labs 2025a) for the six LOOP-areas for eight and a half years from the beginning of 2017 to the middle of 2025. From the Sentinel 2 (optical collection)(PlanetLabs 2025c), 13 bands were downloaded, 12 of them of reflectance data of from wavelength 0.490  $\mu\text{m}$  (blue) to 2.190  $\mu\text{m}$  (short wave infrared). The last band from the Sentinel 2-collection was a scene classification layer (SCL), which was later used in removing pixels affected by clouds, snow or cloud shadows. From Sentinel 1 (radar collection) (PlanetLabs 2025b) two bands were downloaded, the VV and the VH bands.

It is worth noticing that the radar data obtained was equally distributed across the year (fig. 12), but the optical data was available at a much higher frequency during summer than during autumn and winter, see (fig. 13).

Then cutouts the full satellite image data were assigned to field units based on the yearly field polygon layers (Landbrugsstyrelsen 2025). Each observation was made up of data from the 15th of March in the year associated with the field polygon to the 14th of March in the following year. Field-level data was further split into 80 meter \* 80 meter non-overlapping squares as samples to provide more training data, to represent a wider variety of growth patterns for a single crop type, and provide sub-fields of equal size. After filtering out fields with main crops or autumn vegetation types of less than 500 tiles present in the dataset, we were left with 54025 tiles distributed on 5474 different fields. As the size of this data was impractical and unfeasible to handle using free resources, the data was downsampled to be composed of the channel mean in a half-month time window, by the process: For each channelwise time series, outliers were removed, then linear interpolation to daily scale was performed, then the median



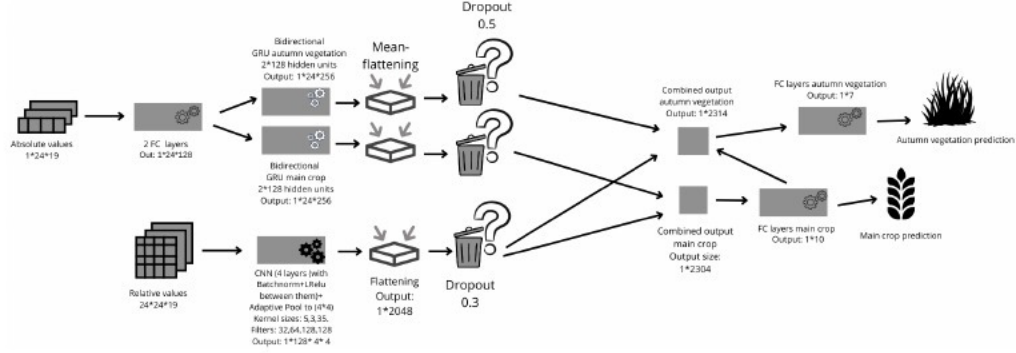


Figure 1: General architecture of the model used.

of the values of each half-month was calculated to represent the period.

The data was then processed into two distinct representations of each field square: A difference-based  $24 \times 24 \times 19$  (channels) representation given by the signed difference between half-monthly means, and an absolute half-monthly mean value representation ( $1 \times 24 \times 19$  (channels)). Values in each representation were normalized by dividing by the range between p5 and p95 of channelwise half-monthly values calculated based on 10.000 random field tiles (taken from the training data to avoid data leakage). The difference matrix is an attempt to represent the temporal patterns of the data in a texture-like manner- something that CNNs are good at learning from. The absolute (but normalized) values were also included as input to the model as a more pure representation of the raw data.

Management and crop-data for the fields in the LOOP-areas was obtained from internal databases at Ecoscience.

## 2.2 Crop categories

Danish agriculture includes a wide variety of crop and catch crop types, which were grouped into broader categories to balance biological and agricultural meaning with sufficient data for model training. Collapsing very specific, unevenly represented categories is inherently imperfect. For this study, main crops were divided into 10 categories and autumn vegetation into 7 (**figs. 14, 15**), which tries to hit a reasonable compromise between detail and sample size.

## 2.3 Model and its exploration

The model (**fig.1**) uses a CNN to process the temporal difference matrix, and two separate GRUs that process the absolute reflectance tensor (one GRU for the main crop and one for autumn vegetation) to learn temporal patterns. The CNN and main crop-GRU features (the means of the hidden states over the

period) are combined to predict the main crop class in an FC-layer, and this main prediction, together with the autumn vegetation-GRU features and CNN-features, is fed into a fully connected layer to classify the autumn vegetation. Like this, the autumn vegetation prediction gets information about the main crop for using in its own prediction. Dropout and normalization layers are applied throughout to improve stability and reduce overfitting.

### 2.3.1 Model training

The loss scheme used was cross entropy loss, and the loss of the autumn vegetation classification task was weighted 10% higher for backpropagation than the main crop classification loss, as to prioritize autumn performance higher.

Training began at learning rate (LR)  $1e-3$  and was reduced by a factor of 0.5 every time summed validation loss for both tasks stagnated for three epochs in a row, after which it fell as low as  $1.5625e-5$  in some runs. Early stopping would have stopped the training if the loss had stagnated for 10 epochs, but this did not happen in practice.

Semi-balanced sampling during model training was performed according to the group’s inverse frequencies in the training data. Half of a batch of 8 was chosen based on the main crop category inverse frequencies and the other half based on the autumn vegetation group inverse frequencies

Very conservative data augmentation was performed out of a fear of destroying the delicate structure of especially the difference matrix; the only distortion of the training data was an added Gaussian Noise to every pixel with standard deviation of 0.02.

### 2.3.2 Combination of model sub-parts in training

8 different modes of integration of the processing of the relative difference images and absolute values images were explored. With the LOOP4-area as the holdout for test, a model was trained for 15-20 epochs (depending on setup), and the result from each run led to the next choice of model. The modes were can be seen in the appendix section 6.4.

Because of these experiments, the final chosen mode was 2 epochs of training the CNN (GRU frozen), 60 epochs of training the GRU (CNN frozen). This was based on the observation that models performed well when either the CNN or the GRU was active (unfrozen) while the other remained frozen. This would mean that half of the model was acting as a extractor extracting but also a regularizer. To emulate that on a higher-epoche scale I chose to allow the CNN to be trained a little bit, before it was frozen, after which it was used as a frozen regularizing feature extractor.

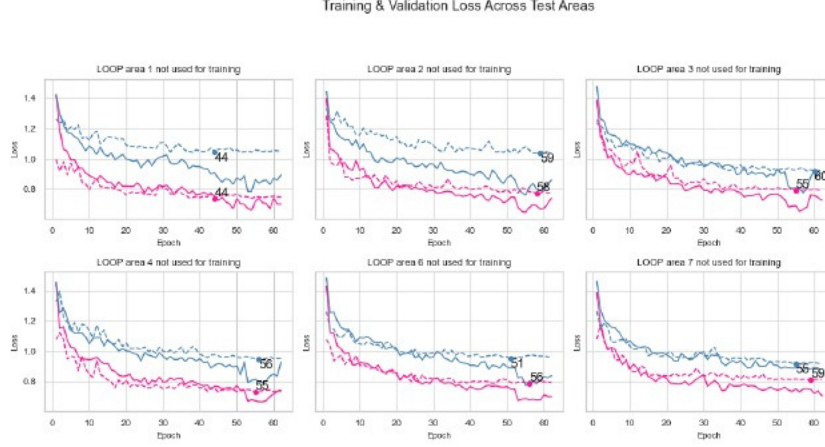


Figure 2: Loss curves from the trainings. In each sub-plot a different of the 6 LOOP-areas was a holdout for later testing of spatial transferability. In blue: The autumn vegetation classification loss across epochs. In pink: the main crop classification loss. In solid: the training loss. In dotted: the validation loss. Annotated: epoch of minimum validation loss.

## 2.4 Model testing

### 2.4.1 Model testing- unseen areas

A very critical aspect of making a generally applicable model is to ensure spatial transferability and generalization. The final model setup was trained in a sort of leave-one-out-cross-validation-scheme on observations from five of the six LOOP-areas and evaluated on the left-out LOOP-area. This means that we have six different estimates on how well the model setup performs on data from unseen areas.

## 3 Results

### 3.1 Training dynamics

In **fig. 2** the loss dynamics during training can be seen for the 6 distinct training runs each holding out a different area to be used as a training set afterwards. All six runs see a steady decrease of loss until loss stagnates or almost stagnates towards the end. In all runs the main crop loss is markedly smaller than the autumn vegetation and it generally reaches low levels using fewer epochs. It is remarkable that the loss is always lower for main crop classification than for autumn vegetation classification because there are more main crop groups, so prediction should be harder by default.

### 3.2 Architecture and training experiments

#### Top 3 Models by Average Accuracy:

- **1st: Mode 5:** (15 epochs GRU, CNN frozen)| *Average Accuracy: 0.811 (Main Crop: 0.906, Autumn Veg.: 0.716)*
- **2nd: Mode 1:**(GRU & CNN trained simult.)| *Average Accuracy: 0.802 (Main Crop: 0.906, Autumn Veg.: 0.698)*
- **3rd: Mode 4:**(15 epochs CNN, GRU frozen)| *Average Accuracy: 0.7982 (Main Crop: 0.8664, Autumn Veg.: 0.730)*

Mode 5 was regarded as the best training mode because after a short exploratory training, it had produced a model with the highest accuracy on the two tasks. Although mode 4 had a higher accuracy in classifying autumn vegetation, all-around success was prioritized as a regularizing strategy.

### 3.3 Test: Generalisation to new areas

As noted in the introduction, the model’s most important desired quality would be its ability to generalize to field tiles from unseen areas. Performance generally improved with larger category sizes (fig. 5), for both autumn vegetation (fig. 8 (in appendix)) and especially main crops (fig. 9 (in appendix)). The model struggled with categories that combined high internal variability and low sample size-particularly “Fallow or none” and “Legumes.” Other groups suffered from either heterogeneity or limited data, but not both; the poor performance on potatoes is likely due to sample scarcity. Overall, the model reached an average F1-score of 0.86 (0.712 unweighted) for main crops and 0.69 (0.685 unweighted) for autumn vegetation (see also fig. 3 and 4).

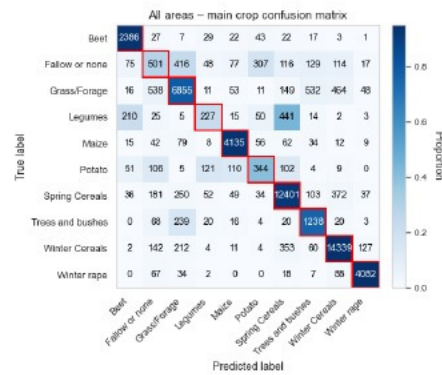


Figure 3: Confusion matrix for main crop classification for unseen areas

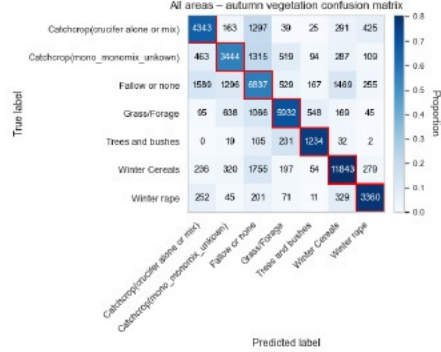


Figure 4: Confusion matrix for autumn vegetation classification for unseen areas

## 4 Discussion

My model was somewhat able to classify reflectance- and radar-data time series into main crop categories, indicating the setup was appropriate. However, the shorter growth period of autumn vegetation (Aug–Nov) produces smaller plants with less distinct patterns, and frequent cloud cover limits optical data, making time series sparse and heavily reliant on interpolations.

As shown in **fig. 5**, crop categories with higher representation achieved higher F1-scores. Despite attempted balanced sampling, the model struggled to generalize for smaller groups, maybe due to limited representation of growth patterns in the training set, which is a natural limitation for heterogeneous categories such as “Fallow or none”.

The conservative use of data augmentation may have limited the model’s ability to learn general representations. Exploring stronger augmentation, focal loss, including main crop information, or passing the GRU’s last hidden states could improve performance.

There may be a natural ceiling to classification accuracy due to limited spatial and temporal resolution of satellite data, the tradeoff between intra-group homogeneity and sample size, and the fact that fields in different categories can appear nearly identical. For example, a grass field may look exactly like a fallow field.

I limited training of my main model to 62 epochs due to computational and time constraints, but exploratory 17-20-epoch runs (see **sec. 3.2**) showed similar test performance (although there, only LOOP 4 was tested as a hold-out), suggesting this did not substantially worsen the model.

## 5 Conclusion

It was generally possible to train my combination model to classify main crops and autumn vegetation types using our rich LOOP-area registration data. However, there are many complicating factors intrinsic to the task: satellite data having limited spatial and temporal details, the hard tradeoff between group homogeneity and sample size, and just simply that fields in different register categories may be highly similar or even indistinguishable.

There may have been complicating factors caused by my own choices: choosing to pool spatial data into field-tile-wide means, the decision to use field tiles and their 80\*80 metre size, the outlier removal can have removed signal, the linear interpolation may have been too simplistic ect.

However, the model attained an average F1-score of 0.86 (weighted by group frequency)/0.712 (unweighted) for main crops and an average F1 of 0.69 (weighted



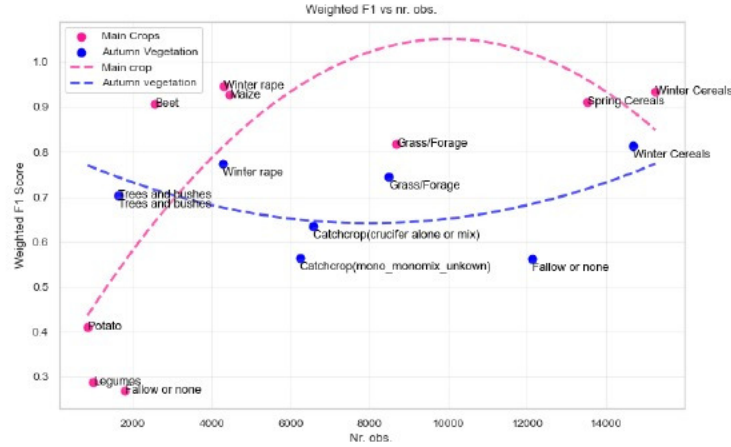


Figure 5: Main crop and autumn vegetation F1 vs. number of observations (unseen areas)

by group frequency)/0.685 (unweighted) for autumn vegetation for field tiles in areas unseen during training, which is reasonable given the challenges.

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## 6 Appendix

### 6.1 Main crop and autumn vegetation categories

|                       |                                                                                                                                                   |                          |                                                                                               |
|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|-----------------------------------------------------------------------------------------------|
| <b>Beet</b>           | Sugar Beets, a very uniform category.                                                                                                             | <b>Potatoes</b>          | Potatoes for different end uses. A homogeneous but small category.                            |
| <b>Fallow or None</b> | Used for explicitly fallow areas, land taken out of production, or fields with no main crop registered in that year, very heterogeneous category. | <b>Spring Cereals</b>    | Including several cereal species with varying cultivation practices. Decently homogeneous.    |
| <b>Grass/Forage</b>   | Including grass with or without clover in varying proportions, whether grazed or cut for hay and/or silage. A somewhat heterogeneous category.    | <b>Trees and Bushes</b>  | Including forest, berry shrubs, orchards, and energy-crop willow, a very heterogeneous group. |
| <b>Legumes</b>        | Encompassing crops such as peas, lupin, faba beans, and others. A very heterogeneous category.                                                    | <b>Winter Cereals</b>    | Comprising different cereal species sown in autumn, very homogenous group.                    |
| <b>Maize</b>          | Covering different cultivation practices and end uses, but generally homogeneous.                                                                 | <b>Winter Rape(seed)</b> | Autumn sown oilseed rape. Very homogeneous group.                                             |

Figure 6: Main crop category overview

|                                                         |                                                                                                                                                                                                                                 |                          |                                                                                               |
|---------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|-----------------------------------------------------------------------------------------------|
| <b>Catch crop (crucifer alone or mix)</b>               | Catch crops that are comprised fully or partly of crucifers such as oil radish and mustard.                                                                                                                                     | <b>Trees and Bushes</b>  | Including forest, berry shrubs, orchards, and energy-crop willow, a very heterogeneous group. |
| <b>Catch crop (monocots, monocot mixes and unknown)</b> | Catch crops that are not known to comprise crucifers, either fully or partially. In practice grasses, cereals (monocots), alone or in mixes with non-crucifers such as Phacelia. Also includes further unspecified catch crops. | <b>Winter Cereals</b>    | Comprising different cereal species sown in autumn, very homogenous group.                    |
| <b>Fallow or None</b>                                   | Used for explicitly fallow areas, land taken out of production, or fields with no main crop registered in that year, very heterogeneous category.                                                                               | <b>Winter rape(seed)</b> | Autumn sown oilseed rape. Very homogeneous group.                                             |
| <b>Grass/Forage</b>                                     | Including grass with or without clover in varying proportions, whether grazed or cut for hay and/or silage. A somewhat heterogeneous category.                                                                                  |                          |                                                                                               |

Figure 7: Autumn vegetation category overview

## 6.2 F1 values vs observation numbers for unseen areas (details)

| Autumn vegetation |                                  |             |          |
|-------------------|----------------------------------|-------------|----------|
|                   | Crop category                    | Weighted F1 | Nr. obs. |
| 0                 | Catchcrop(crucifer alone or mix) | 0.635       | 6583     |
| 1                 | Catchcrop(mono_monomix_unkown)   | 0.565       | 6231     |
| 2                 | Fallow or none                   | 0.562       | 12142    |
| 3                 | Grass/Forage                     | 0.745       | 8493     |
| 4                 | Trees and bushes                 | 0.704       | 1623     |
| 5                 | Winter Cereals                   | 0.813       | 14684    |
| 6                 | Winter rape                      | 0.774       | 4269     |

Figure 8: Autumn vegetation F1 vs. number of observations (unseen areas)

| Main crops |                  |             |          |
|------------|------------------|-------------|----------|
|            | Crop category    | Weighted F1 | Nr. obs. |
| 0          | Beet             | 0.908       | 2557     |
| 1          | Fallow or none   | 0.269       | 1800     |
| 2          | Grass/Forage     | 0.818       | 8677     |
| 3          | Legumes          | 0.288       | 992      |
| 4          | Maize            | 0.927       | 4452     |
| 5          | Potato           | 0.412       | 852      |
| 6          | Spring Cereals   | 0.911       | 13515    |
| 7          | Trees and bushes | 0.704       | 1628     |
| 8          | Winter Cereals   | 0.935       | 15254    |
| 9          | Winter rape      | 0.947       | 4298     |

Figure 9: Main crop F1 vs. number of observations (unseen areas)

## 6.3 Crop composition and difference matrix examples

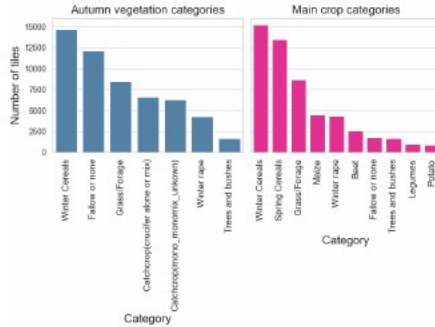


Figure 10: Overview of main crop and autumn vegetation category frequencies after removing groups with less than 500 tiles

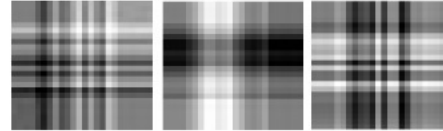


Figure 11: Three relative difference matrix examples. Each pixel compares one half-month period to another (for one band).

## 6.4 Model exploration modes

- Mode 1: 15 epochs, neither CNN nor GRU frozen.
- Mode 2: 10 epochs CNN (GRU frozen), 10 epochs GRU (CNN frozen).
- Mode 3: 10 epochs GRU (CNN frozen), 10 epochs CNN (GRU frozen).
- Mode 4: 15 epochs CNN only (GRU frozen but present).
- Mode 5: 15 epochs GRU only (CNN frozen but present).



- Mode 6: 15 epochs CNN (no frozen GRU).
- Mode 7: 15 epochs GRU (no frozen CNN).

## 6.5 Median observations available radar and optical

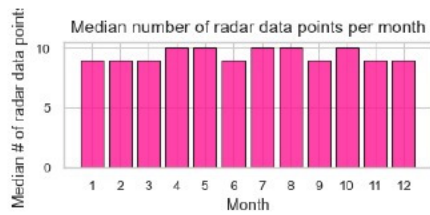


Figure 12: Median valid number of radar data acquisitions for each month of the year for 2000 randomly selected field-tiles (before additional outlier removal).

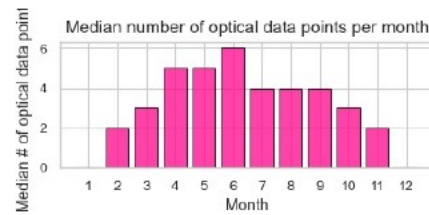


Figure 13: Median valid number of optical data acquisitions for each month of the year for 2000 randomly selected field-tiles (before additional outlier removal).

## 6.6 Code

See [https://github.com/vegetation-classification/DPL\\_proj\\_vegetation\\_classification](https://github.com/vegetation-classification/DPL_proj_vegetation_classification)